Using First and Second Derivatives

Preliminaries: Previous facts about f, f', and f''

- If $f' > 0$ on an interval, then f is **increasing** on that interval.
- If $f' < 0$ on an interval, then f is **decreasing** on that interval.
- If $f'' > 0$ on an interval, then f is **concave up** on that interval.
- If $f'' < 0$ on an interval, then f is **concave down** on that interval.

The First Derivative Test

Let a be a point in the domain of a function f. Then:

- f has a **local maximum** at a if $f(a) \geq f(x)$ for all x values near a.
- f has a **local minimum** at a if $f(a) \leq f(x)$ for all x values near a.
- a is called a **critical point** of f if $f'(a) = \underline{0}$ or $f'(a)$ is **undefined** (or DNE).

*Note: We use the term **extrema** to refer to both maxima and minima.*

Question: Consider the graphs below:

Case A (Valley / Local Minimum)	**Case B (Peak / Local Maximum)**
Slope goes from downhill to uphill: $f' < 0 \quad (-)$ before x_0 $f' > 0 \quad (+)$ after x_0	Slope goes from uphill to downhill: $f' > 0 \quad (+)$ before x_0 $f' < 0 \quad (-)$ after x_0

What happens to f' around x_0 in each of the graphs?

A: As we pass x_0, f' goes from **negative to positive** $(- \to +)$. **(Valley / Local Minimum)**

B: As we pass x_0, f' goes from **positive to negative** $(+ \to -)$. **(Peak / Local Maximum)**

The First Derivative Test (Summary Statement)

Let x_0 be a critical point of a continuous function f:

- If f' changes sign from **positive to negative** $(+ \to -)$, then f has a **local max** at x_0.
- If f' changes sign from **negative to positive** $(- \to +)$, then f has a **local min** at x_0.
- (If f' does not change sign across x_0, then f has **neither** a local max nor a local min at x_0).

Using First and Second Derivatives

Question: Find and classify all of the local extrema of $f(x) = x^3 - 3x + 4$. (i.e. decide whether they are local maxima, minima, or neither.)

Step 1: Derivative: $f'(x) = 3x^2 - 3$

Step 2: Critical Points: $f'(x) = 0 \implies 3x^2 - 3 = 0 \implies 3x^2 = 3 \implies x^2 = 1 \implies x = \pm 1$.

Step 3: First Derivative Sign Chart (Global Parabola View):

x:	$(-\infty, -1)$	-1	$(-1, 1)$	1	$(1, \infty)$
Sign of $f'(x)$:	$+$	0	$-$	0	$+$
Behavior of f:	Increasing	Peak	Decreasing	Valley	Increasing

Step 4: Classification via First Derivative Test:

- At $x = -1$, f' goes from $+$ to $-$ $\implies$ **local max at** $x = -1$ with $f(-1) = 6$.
- At $x = 1$, f' goes from $-$ to $+$ $\implies$ **local min at** $x = 1$ with $f(1) = 2$.

Question: True or False?: If $f'(x_0) = 0$ then f has a local max or min at x_0. Explain.

Answer: False.
Explanation / Counterexample: Let $f(x) = x^3$, then $f'(x) = 3x^2$. Here $3x^2 = 0$ at $x = 0$ (so $x = 0$ is a critical point). However, $3x^2 \geq 0$ for all x, meaning $f'(x) > 0$ both before and after 0 ($+ \to +$). The function flattens out at 0 but keeps increasing. Therefore, $x = 0$ is neither a local maximum nor a local minimum.

The Second Derivative Test

Consider again the plots on the previous page. How does the concavity of a function at a critical point relate to local extrema?

- **Local Min (Valley):** Concave Up ($f'' > 0$, smile $\cup$)
- **Local Max (Peak):** Concave Down ($f'' < 0$, frown $\cap$)

The Second Derivative Test Statement:

- If $f'(x_0) = 0$ and $f''(x_0) > 0$, then f has a **local minimum** at x_0.
- If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a **local maximum** at x_0.
- If $f'(x_0) = 0$ and $f''(x_0) = 0$, then the test **gives no information (fails)**.

Question: Use the second derivative test to classify the critical points of $f(x) = x^3 - 3x + 4$.

- From above, $f'(x) = 3x^2 - 3 \implies$ Critical points at $x = \pm 1$.
- Second derivative: $f''(x) = 6x$.
- At $x = -1$: $f''(-1) = 6(-1) = -6 < 0 \implies$ **local max ($L_{\max}$) at** $x = -1$.

- At $x = 1$: $f''(1) = 6(1) = 6 > 0 \implies$ **local min (L_{min}) at** $x = 1$.

Using First and Second Derivatives

Question: Classify the critical point $x = 0$ for $y = x^3$, $y = x^4$, and $y = -x^4$.

1) For $y = x^3$:

- $f'(x) = 3x^2 \implies$ critical point at $x = \underline{0}$.
- Second derivative: $f''(x) = 6x \implies f''(0) = 0 \implies$ **2nd Derivative Test FAILS!**
- First Derivative Test: Sign of f':

$$(-\infty, 0): \ (+) \quad \longrightarrow \quad (0, \infty): \ (+)$$

Sign of f' goes from $+$ to $+$ (does not change) $\implies x = 0$ is **NOT a local min or max**.

2) For $y = x^4$:

- $f'(x) = 4x^3 \implies$ critical point at $x = 0$.
- Second derivative: $f''(x) = 12x^2 \implies f''(0) = 0 \implies$ **2nd Derivative Test FAILS!**
- First Derivative Test: Sign of f':

$$(-\infty, 0): \ (-) \quad \longrightarrow \quad (0, \infty): \ (+)$$

Sign of f' goes from $-$ to $+ \implies x = 0$ is a **local minimum (L_{min})**.

3) For $y = -x^4$:

- $f'(x) = -4x^3 \implies$ critical point at $x = 0$.
- Second derivative: $f''(x) = -12x^2 \implies f''(0) = 0 \implies$ **2nd Derivative Test FAILS!**
- First Derivative Test: Sign of f':

$$(-\infty, 0): \ (+) \quad \longrightarrow \quad (0, \infty): \ (-)$$

Sign of f' goes from $+$ to $- \implies x = 0$ is a **local maximum (L_{max})**.

Inflection Points

Definition: A function f has an **inflection point** at a if the **concavity of f changes** at a.
At an inflection point x_0, $f''(x_0) = 0$ or $f''(x_0)$ is **undefined** (or DNE).

WARNING!!! Not every point where $f''(x) = 0$ is an inflection point (just as not every point where $f'(x) = 0$ is a local max or local min). Example?

Example: $f(x) = x^4$ at $x = 0$.
Here $f''(x) = 12x^2$, so $f''(0) = 0$. But for all $x \neq 0$, $12x^2 > 0$ (concave up on both sides). Because the concavity does not change sign across 0, $x = 0$ is **NOT an inflection point** (it is a local minimum).